WORKED SOLUTION

STEP I 2004, Question 11

Mechanics • Friction

STEP I 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q11

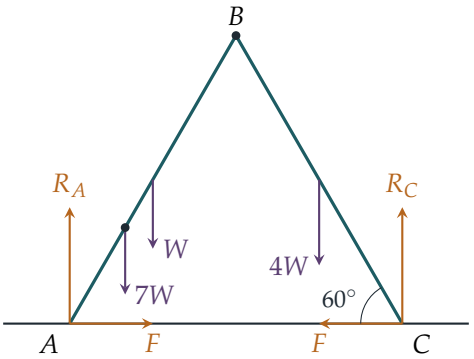
The key idea

Before anything slips, the whole frame is in equilibrium, so we can find both normal reactions from the whole system and the friction from one ladder on its own (taking moments about the hinge removes the unknown hinge force). The only horizontal external forces are the two frictional forces, so they are equal in size. Whichever foot has the *smaller* normal reaction needs the larger ratio F/R, and that foot is the one that goes first.

Solution

We model each ladder as a uniform rod of length $2a$, the hinge at B as smooth, and the decorator as a particle. Since $AB = BC$ and $\angle ABC = 60°$, triangle ABC is equilateral: $AC = 2a$ and each ladder makes $60°$ with the ground. Then B is a horizontal distance a from each foot and a height $a\sqrt{3}$ above the ground.

When the decorator is a third of the way up AB, she is a horizontal distance $\frac{1}{3}a$ from A. The centre of AB is $\frac{1}{2}a$ from A, and the centre of BC is $\frac{3}{2}a$ from A.



Normal reactions

Let the normal reactions at A and C be R_A and R_C. Resolving vertically for the whole system,

$$R_A + R_C = W + 4W + 7W = 12W.$$

Taking moments about A for the whole system (the hinge force is internal, so it does not appear):

$$R_C \cdot 2a = W \cdot \tfrac{1}{2}a + 7W \cdot \tfrac{1}{3}a + 4W \cdot \tfrac{3}{2}a = \tfrac{53}{6}Wa,$$

so

$$R_C = \tfrac{53}{12}W, \qquad R_A = 12W - \tfrac{53}{12}W = \tfrac{91}{12}W.$$

Friction

The feet tend to splay apart, so friction at each foot points inwards. Resolving horizontally for the whole system, the frictional forces are the only horizontal external forces, so they have the same magnitude; call it F.

To find F we need one ladder on its own. Take BC, which carries no decorator, and take moments about B so that the unknown hinge force drops out. About B, the reaction R_C has lever arm a, the weight $4W$ has lever arm $\frac{1}{2}a$, and the friction F (horizontal, at ground level) has lever arm $a\sqrt{3}$. The weight and F turn BC one way and R_C turns it the other, so

$$R_C \cdot a = 4W \cdot \tfrac{1}{2}a + F \cdot a\sqrt{3} \quad \Longrightarrow \quad F = \frac{R_C - 2W}{\sqrt{3}} = \frac{29W}{12\sqrt{3}}.$$

As a check, the same calculation for AB about B gives $R_A \cdot a = W \cdot \frac{1}{2}a + 7W \cdot \frac{2}{3}a + Fa\sqrt{3}$, so $F\sqrt{3} = \frac{91}{12}W - \frac{31}{6}W = \frac{29}{12}W$, which agrees.

Which ladder slips?

> **Ask yourself** The friction is the same at both feet, so how can we tell which one gives way?

A foot is on the point of slipping when the friction needed equals μ times the normal reaction there. Both feet need the same F, so the foot with the smaller normal reaction needs the larger ratio:

$$\frac{F}{R_A} = \frac{29}{91\sqrt{3}} \approx 0.184, \qquad \frac{F}{R_C} = \frac{29}{53\sqrt{3}} \approx 0.316.$$

So C reaches limiting friction first: it is the **heavier ladder** BC that slips, even though the decorator is on AB. Her weight presses A firmly into the ground, while C carries much less of the load.

We should also make sure nothing slipped earlier. If she is a fraction t of the way up,

the same working gives $R_C = (\frac{13}{4} + \frac{7}{2}t)W$ and $F\sqrt{3} = R_C - 2W = (\frac{5}{4} + \frac{7}{2}t)W$, so

$$\frac{F}{R_C} = \frac{1}{\sqrt{3}} \cdot \frac{\frac{5}{4} + \frac{7}{2}t}{\frac{13}{4} + \frac{7}{2}t} = \frac{1}{\sqrt{3}} \left(1 - \frac{2}{\frac{13}{4} + \frac{7}{2}t}\right),$$

which increases with t. Similarly F/R_A increases with t (its numerator rises and its denominator $R_A = 12W - R_C$ falls). So the required ratios grow steadily as she climbs, and the first slip happens exactly when F/R_C reaches μ, at $t = \frac{1}{3}$. Hence

$$\mu = \frac{29}{53\sqrt{3}} = \frac{29\sqrt{3}}{159} \approx 0.316.$$

Checks

- Moments for AB about B give the same F as moments for BC about B (shown above), so the reactions are consistent.
- With no decorator at all, R_C should exceed R_A because BC is heavier; indeed moments about A then give $R_C = \frac{13}{4}W$ and $R_A = \frac{7}{4}W$, adding to $5W$.
- $R_A = R_C$ when $t = \frac{11}{14}$; only beyond that point would AB be the one to slip, so at $t = \frac{1}{3}$ it must be BC.
- Numerically, recomputing every force from coordinates (and checking moments about B for both ladders) gives $\mu \approx 0.31591$.

Common slips

- Assuming the ladder carrying the decorator slips. Her weight increases the normal reaction at A, making that foot *more* secure.
- Taking moments about a foot for a single ladder, which brings in the unknown hinge force. Moments about B for one ladder avoid it.
- Forgetting that the friction has lever arm $a\sqrt{3}$ (the height of B) about the hinge.

Beyond STEP

This is the basic strategy of statics for jointed structures: treat the whole structure to find external reactions, then cut it at a joint to find what one member must supply. Questions of which contact fails first lead on to the analysis of frames and to the principle of virtual work, where equilibrium is found by considering small displacements consistent with the constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

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